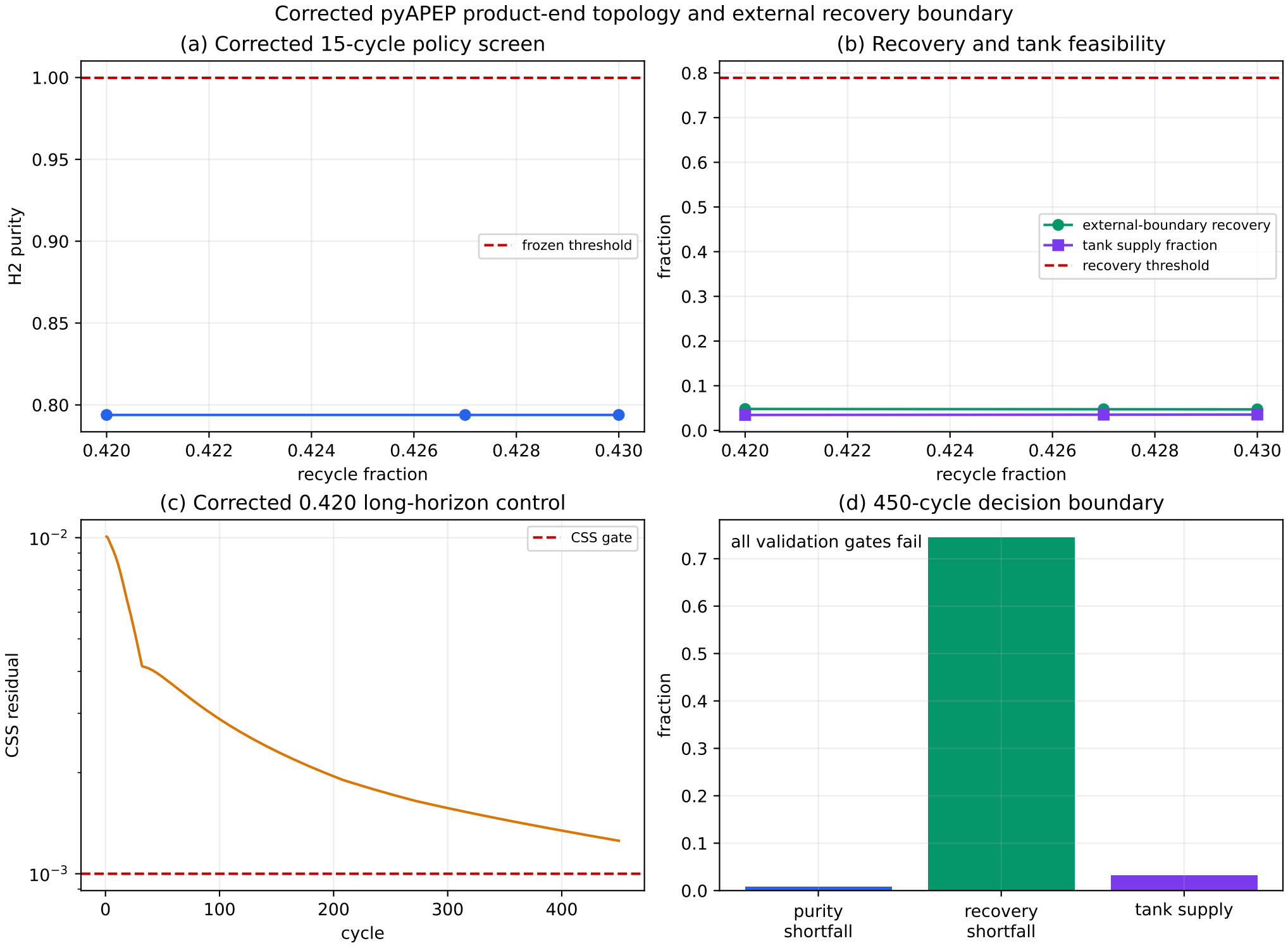
Corrected pyAPEP product-end topology and external recovery boundary
(a) Corrected 15-cycle policy screen
H2 purity
1.00
0.95
0.90
0.85
0.80
frozen threshold
0.420
0.422
0.424
0.426
0.428
0.430
recycle fraction
(b) Recovery and tank feasibility
fraction
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
external-boundary recovery
tank supply fraction
recovery threshold
0.420
0.422
0.424
0.426
0.428
0.430
recycle fraction
(c) Corrected 0.420 long-horizon control
CSS residual
10^{-2}
10^{-3}
CSS gate
0
100
200
300
400
cycle
(d) 450-cycle decision boundary
fraction
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
all validation gates fail
purity shortfall
recovery shortfall
tank supply